

JURONG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATIONS  
Higher 2

CANDIDATE  
NAME

CLASS

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## BIOLOGY

**9648/03**

Applications Paper and Planning Question

**16 September 2013**

Paper 3

**2 hours**

Additional Materials: Answer Booklet/Paper

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### READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

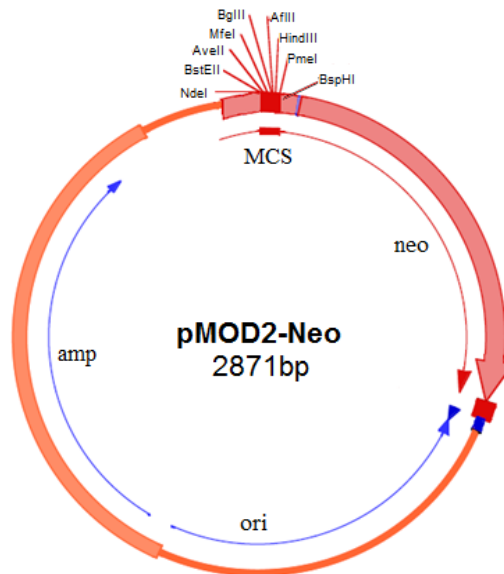
Answer **all** questions.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
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| 1                  |  |
| 2                  |  |
| 3                  |  |
| 4                  |  |
| 5                  |  |
| Total              |  |

Answer **all** questions.

- 1 Plasmids used in genetic engineering as a cloning vector include pMOD2-Neo shown in Fig. 1.1. This plasmid contains genes coding for antibiotic resistance – ampicillin resistance gene (amp) and neomycin resistance gene (neo).



**Fig. 1.1**

With reference to the plasmid shown in Fig. 1.1,

- (a) State two properties of the plasmid which allows it to be used as a cloning vector. [2]

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- (b) The plasmid pMOD2-Neo and the human DNA containing the gene of interest are cut with restriction enzymes and cloned using bacteria in genetic engineering.

- (i) Explain what is meant by a *restriction enzyme*. [2]

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- (ii) Suggest why the restriction enzyme will cut the human DNA in many places but will only cut the plasmid DNA once. [1]

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- (iii) Suggest why the plasmid should only be cut once. [1]

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- (c) Disease X is an autosomal disease resulting from a single base substitution which removes a recognition site of the restriction enzyme, *HaeIII*, as shown in Fig. 1.2. The enzyme's recognition sites on allele 1 and allele 2 are indicated by arrows.

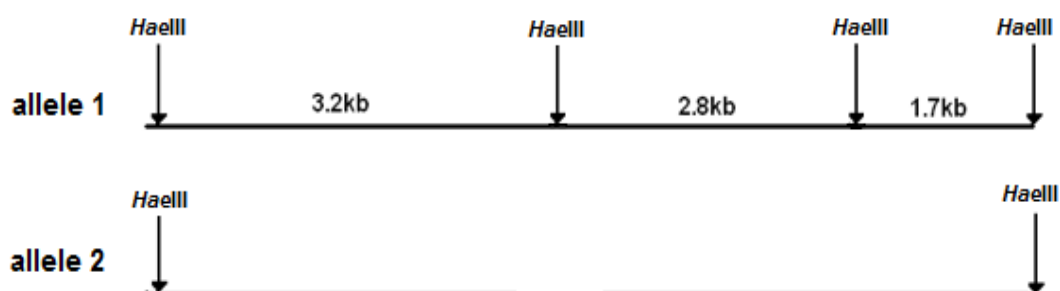


Fig. 1.2

The mutation can be detected by restriction fragment length polymorphism after cleaving with *HaeIII*. Fig. 1.3 shows the banding patterns produced by four individuals.

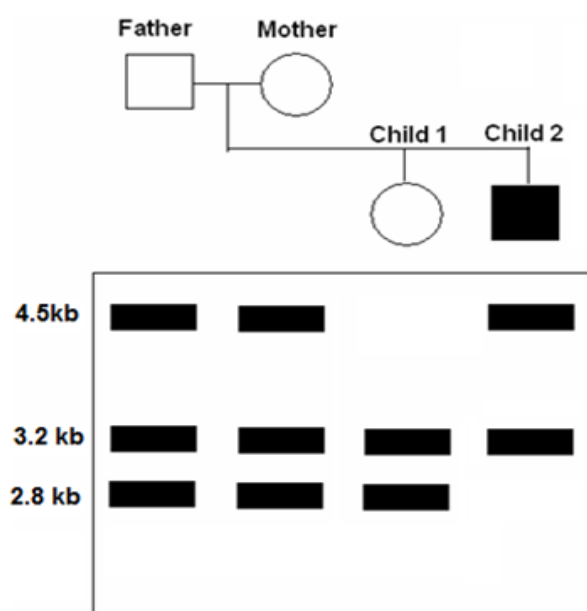


Fig. 1.3

With reference to Fig. 1.2 and Fig. 1.3,

- (i) On allele 1, draw the probe at the position it will hybridise to. [1]
- (ii) On allele 2, draw the position of the polymorphic restriction site by marking it with an asterisk (\*). [1]
- (iii) State the mode of inheritance for Disease X. [1]

**(iv)** Explain how gel electrophoresis can be used to distinguish between the two alleles. [4]

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**(d)** Information from the Human Genome Project (HGP) has allowed for better understanding of the genetic basis of diseases such as Disease X.

Discuss two other benefits arising from HGP. [2]

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[Total: 15]

- 2 Haemophilia B is a blood clotting disorder. Haemophilia B patients are unable to produce enough human clotting factor IX, also referred to as FIX, which is an essential protein in the blood clotting process.

The dominant allele coding for FIX was packaged in a modified adeno-associated virus, AAV8, that has been designed to infect liver cells specifically. The modified AAV8 was injected into six patients with severe haemophilia B who were producing FIX at less than 1% of normal levels.

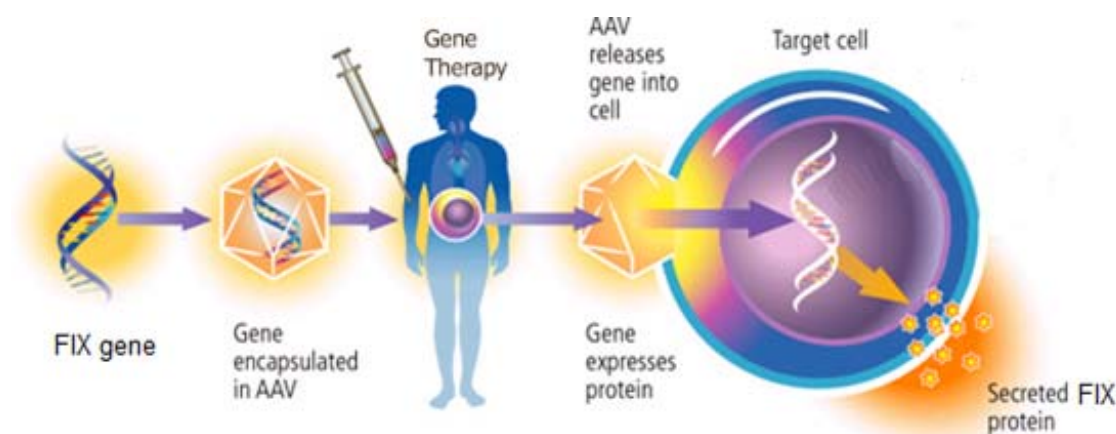


Fig. 2.1

Prior to undergoing gene therapy, all six patients were receiving the standard treatment for severe haemophilia B— injection of purified FIX produced by recombinant DNA technology—several times a month. After receiving the FIX gene therapy, each patient produced FIX at between 2% to 12% of normal levels. For at least six months, four of the six patients no longer required purified FIX injections for routine bleeding. The other two patients required purified FIX injections less frequently than before the therapy.

- (a) Suggest and explain why two out of the six patients still require purified FIX injection. [2]

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- (b) Explain why it is sufficient to insert a single dominant allele coding for FIX into each liver cell. [2]

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- (c) Explain why individuals who have been treated by this method of gene therapy do not pass on the dominant allele coding for FIX to their children. [1]

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- (d) State two disadvantages of this type of gene therapy. [2]

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- (e) Suggest and explain the advantages of using a viral delivery system for the dominant allele coding for FIX. [2]

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[Total: 9]

- 3 Investigators often report success in culturing plant tissue using root tips. This technique has many commercial applications which include the production of large numbers of genetically identical plants and the culturing of plant cells in a fermenter with culture medium for the production of secondary products.

(a) (i) Suggest one reason why plant tissue culture was successful using organs such as root tips. [1]

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(ii) Suggest why it is advantageous that plants produced by plant tissue culture are genetically identical. [1]

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In many stages of plant tissue culture, strict aseptic techniques need to be used. One of the difficulties of culturing plant tissue is in carrying out surface sterilisation of the explant.

To investigate a suitable method of culturing explants, plant tissue samples were taken at three different times of the year and placed on a medium containing no fungicide or antibiotic. Two different sampling methods were used:

- discs cut from leaves wiped with bleach;
- stem tissues extracted with a sterile hypodermic syringe.

The results of culturing the explants are shown in Fig. 3.1.

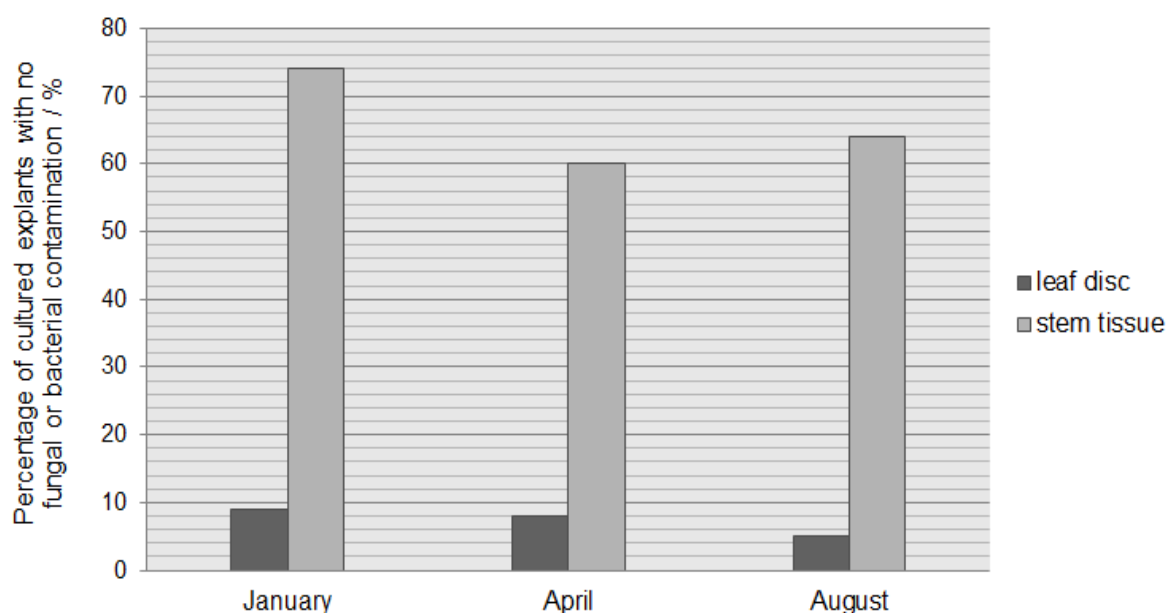


Fig. 3.1

With reference to Fig. 3.1,

- (b)** compare the effect of the different methods of taking tissue samples and the time of the year on fungal or bacterial contamination of the cultured explants. [3]

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- (c) Early attempts to culture plant tissue on a culture medium were not always successful. Studies have shown that the culture medium must contain a range of substances - mineral ions, a sugar suitable for plant cells to use as well as plant growth regulators, to bring about successful plant cell growth and division.

To investigate the role of plant growth regulators, equal masses of plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 3.1.

**Table 3.1**

| treatment | concentration of plant growth regulators / mg dm <sup>-3</sup> |           | effect of plant growth regulators on callus growth |
|-----------|----------------------------------------------------------------|-----------|----------------------------------------------------|
|           | auxin                                                          | cytokinin |                                                    |
| A         | 2.00                                                           | 0.00      | little or no growth                                |
| B         | 2.00                                                           | 0.02      | growth of roots                                    |
| C         | 2.00                                                           | 0.20      | increased growth of callus with no differentiation |
| D         | 2.00                                                           | 0.50      | growth of shoots                                   |
| E         | 0.00                                                           | 0.20      | little or no growth                                |

- (i) Suggest why a sugar was included in the culture medium. [1]

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With reference to Table 3.1,

**(ii)** describe the effects of auxin and cytokinin on callus growth. [3]

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**(iii)** Suggest and explain what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets. [2]

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Plantlets often do not have fully functional cuticle and this causes them to be inefficient in their use of water. Hence, plantlets might not be able to survive the ex-vivo acclimatisation process if directly transferred from the culture medium to the greenhouse.

**(iv)** Suggest what steps need to be taken to increase the viability of the plantlets. [2]

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- (d) In many regions of temperate rice production, severe weed infestations have removed land from economic rice production. LLRICE62 is adapted for planting in the temperate rice production regions.

LLRICE62 is a variety of *Oryza sativa* L. rice produced through genetic modification to confer tolerance to the herbicide glufosinate ammonium. Herbicide tolerance is based upon the *bar* gene, a bialaphos resistance gene, isolated from the soil microorganism, *Streptomyces hygroscopicus*. The *bar* gene encodes the production of the enzyme, Phosphinothricin-Acetyl-Transferase (PAT). When sprayed with glufosinate ammonium herbicides, the LLRICE62 plants can continue to grow while surrounding weeds rapidly die.

Based on the above information, suggest

- (i) the benefit of cultivating LLRICE62. [1]

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- (ii) two ethical implications of this genetic engineering of LLRICE62. [2]

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[Total: 16]

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#### 4 Planning question

You are required to plan, but not carry out, an investigation into the effect of alcohol concentration on the membrane permeability of beetroot tissue. Beetroot is a starchy edible root from the *Beta vulgaris* plant whose cells contain a water-soluble red pigment, betacyanin, in their vacuoles. This pigment cannot pass through membranes unless the membrane is damaged.

Fig. 4.1 shows a spectrophotometer which can be used to measure colour intensity by measuring absorbance of different wavelengths of light as they pass through the sample. A cuvette filled with sample to be placed into the sample holder is also shown. Generally, a sample of higher colour intensity will absorb more light energy at a specific wavelength.

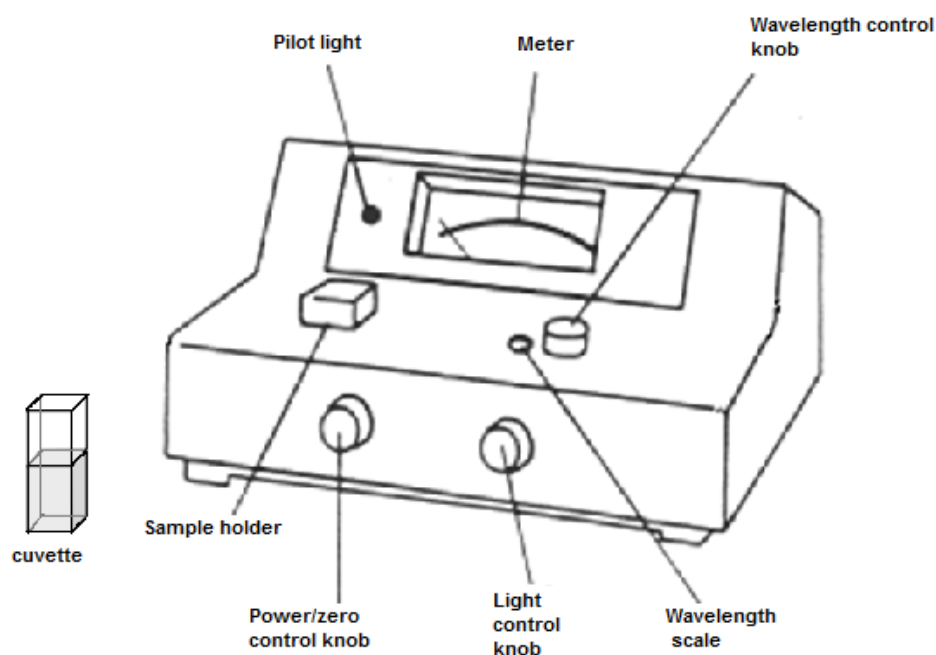


Fig. 4.1

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variable,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

Your planning must be based on the assumption that you have been provided with the following equipment and material, which you must use:

- a large piece of beetroot tissue with the skin removed,
- test tubes – 100mm tall with 11mm inside diameter,
- scalpel,
- 10.0% alcohol,
- Spectrophotometer set at 460nm,
- cuvettes,
- distilled water.

You should select from the following apparatus:

- white tile,
- ruler,
- timer e.g. stopwatch or stopclock,
- any normal laboratory glassware e.g. different sized beakers, measuring cylinders or syringes for measuring volume.

[Total: 12]

This image shows a full page of white paper with horizontal grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.



[illegible]

### Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Describe the polymerase chain reaction (PCR) and explain the advantages of this procedure. [8]

**(b)** Describe, with examples, how a eukaryotic gene can be incorporated into *E.coli*. [6]

**(c)** Explain how RFLP analysis facilitates the process of DNA fingerprinting and genomic mapping. [6]

[Total: 20]